

IMP (Mids)

CLO 1 → 15 Marks

CLO 2 → 15 Marks

CLO 1 (Theory)

Chp 1 Error (Definition of Error Source of Error Type of Error)

- What is error in numerical analysis
- Applications of errors in computer science

CLO 2

Chp 2

Non linear Equation

- 1) Bisection → Imp * 4 iterations fixed
- 2) Regula Falsi
- 3) Secant
- 4) Falsi
- 5) Fixed Point

LU factorization $\rightarrow$ class
~~Jab~~ Jacobian matrix $\rightarrow$ questions

Gradient Decent (1st Order)
Hessian matrix (2nd Order)

Why we use optimizing in numerical analysis?

Optimizing algorithms

- Gradient Descent
- Hessian Matrix

Interpolation (Theory)

What is Interpolation? (First 3 lines)

Why we use

Applications of Interpolation and Types of Interpolation (only headings and subheadings only)

Linear Interpolation

Headings and Subheadings only
↳ headings + conclusion

$$\begin{array}{r} 2 \\ 3 \\ 4 \end{array}$$

Date: _____

Newton Forward } Difference
Newton Backward }
Formula
Applications ~~ap~~

When we have numeric data:-

$$\text{Error} = \text{Exact Value} - \text{Approximate Value}$$

$$f = \text{Exact Value} = \pi$$

$$\text{Approximate Value} = 3.14$$

$$\text{Error} = \pi - 3.14$$

Types of Error:-

Gross Error $\rightarrow$ caused by human mistake
computer
, or ~~human~~ error

When we have variable form:-
 x , $3x^n$

→ derivative of any function

$$\text{Absolute Error} = 3x^2 \cdot \Delta x$$

$$\begin{aligned} A.E &= \Delta y \\ R.E &= \delta y \end{aligned}$$

$$\text{Relative Error} = \frac{\text{Absolute Error}}{\text{Exact Value}}$$

Relative Error
Exact Value

$$= \frac{3x^2 \cdot \Delta x}{x^3}$$

$$\boxed{\text{Relative Error} = \frac{3}{x} \Delta x}$$

$$\text{Absolute Error} = 3x^{n-1}$$

$$\ln(Sx^2 + 9)$$

$$e^{3x^2 + 5x}$$

$$9^x$$

$$e^{3n^2+5n}$$

$$e^{3n^2+5n} \frac{d}{dn} 3n^2+5n$$

$$(e^{3n^2+5n})(6n+5) \cdot \Delta n = \text{Absolute Error}$$

$$\text{Relative Error} = \frac{\text{Absolute Error}}{\text{Exact Value}}$$

$$= \frac{(e^{3n^2+5n})(6n+5) \Delta n}{e^{3n^2+5n}}$$

$$\boxed{\text{Relative Error} = (6n+5) \Delta n}$$

Power Base Angle

$$\ln^7(5x^2+9)$$

$$7 \ln(5x^2+9) \cdot \frac{1}{5x^2+9} \times 10x \rightarrow \text{Absolute Error}$$

$$\boxed{\text{Relative Error} = \frac{7 \ln(5x^2+9) \cdot \frac{1}{5x^2+9} \times 10x}{\ln^7(5x^2+9)}}$$

$$q^n$$

$$y = a^n$$

$$\cancel{q^n} \cdot \frac{d}{dn} \cancel{q^n}$$

$$\frac{\Delta y}{\Delta n} = a^n \ln a$$

$$\cancel{18} \cdot \cancel{n-1} \cdot \cancel{\Delta n}$$

$$q^n \ln q \Delta n, \text{ Absolute Error}$$

$$R.E = \frac{A.E}{E.V}$$

$$= \frac{\cancel{q^n} \ln q \Delta n}{\cancel{q^n}}$$

$$R.E = \ln q \Delta n$$

Relative Error =

$$\frac{\text{Current Value} - \text{Previous Value}}{\text{Current Value}} \times 100$$

Newton's Method (Raphson):-

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Maximum Iterations = 4

Mathews

Secant Method :-

$$x_2 = x_1 - \frac{f(x_1) [x_1 - x_0]}{f(x_1) - f(x_0)}$$

False Position Method :-

$$c = a - \left[\frac{b - a}{f(b) - f(a)} \right] \times f(a)$$

4 Marks / 3 Marks

Fixed point Iteration Method:-

Fixed Point Iteration

$$x^3 + x^2 - 1 = 0$$

if initial value
not given then
initial value = 0

$|\phi'(x)| < 1 \rightarrow$ Optimized equation
(Derivative should be
less than 1)

Case No 1

$$x^3 + x^2 - 1 = 0$$

$$x^3 = 1 - x^2$$

$$(x^3)^{1/3} = (1 - x^2)^{1/3}$$

$$x = (1 - x^2)^{1/3}$$

Case No 2

$$x^3 + x^2 - 1 = 0$$

$$x^2 = 1 - x^3$$

$$(x^2)^{1/2} = (1 - x^3)^{1/2}$$

$$x = (1 - x^3)^{1/2}$$

Case No 3

$$x^3 + x^2 - 1 = 0$$

$$x^2(x+1) = 1$$

$$x^2 = \frac{1}{(x+1)}$$

$$x = \frac{1}{\sqrt{x+1}}$$

$x=2$

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$$\frac{d}{dn}(n) = \frac{d}{dn}(1-n^2)^{\frac{1}{3}}$$

$$= \frac{1}{3}(1-n^2)^{\frac{1}{3}-1} \frac{d}{dn}(1-n^2)$$

$$= \frac{1}{3}(1-n^2)^{-\frac{2}{3}} - 2n$$

$$= \frac{-2n(1-n^2)^{-\frac{2}{3}}}{3}$$

Initial value = 0.5

$$\Rightarrow \left| \frac{-2(0.5)(1-(0.5)^2)^{-\frac{2}{3}}}{3} \right| < 1$$

$$\Rightarrow |-0.403| < 1$$

$$\Rightarrow \underline{\underline{0.403}}$$

$$\frac{d}{dn}(n) = \frac{d}{dn}(1-n^3)^{\frac{1}{2}}$$

$$= \frac{1}{2}(1-n^3)^{\frac{1}{2}-1} \frac{d}{dn}(1-n^3)$$

$$= \frac{1}{2}(1-n^3)^{-\frac{1}{2}} - 3n$$

$$= \frac{-3n(1-n^3)^{-\frac{1}{2}}}{2}$$

Initial value = 0.5

$$\Rightarrow \left| \frac{-3(0.5)(1-(0.5)^3)^{-1/2}}{2} \right| < 1$$

$$\Rightarrow | -0.534 | < 1$$

$$\Rightarrow \underline{\underline{0.534}}$$

$$\begin{aligned} \frac{d}{dx}(x) &= \frac{d}{dn}(x+1)^{-1/2} \\ &= -\frac{1}{2}(x+1)^{-1/2-1} \frac{d}{dn}(x+1) \\ &= -\frac{1}{2}(x+1)^{-3/2} \end{aligned}$$

Initial value = 0.5

$$\Rightarrow \left| \frac{-1}{2}(0.5+1)^{-3/2} \right| < 1$$

$$\Rightarrow | -0.272 | < 1$$

$$\Rightarrow \underline{\underline{0.272}}$$

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Therefore,

$$x_1 = \frac{1}{\sqrt{x+1}}$$

$$= \frac{1}{\sqrt{0.5+1}}$$

$$\boxed{x_1 = 0.8164}$$

$$x_2 = \frac{1}{\sqrt{x_1+1}}$$

$$\rightarrow x_2 = \frac{1}{\sqrt{0.8164+1}}$$

$$\boxed{x_2 = 0.7419}$$

$$x_3 = \frac{1}{\sqrt{x_2+1}}$$

$$x_3 = \frac{1}{\sqrt{0.7419+1}}$$

$$\boxed{x_3 = 0.7576}$$

$$x_4 = \frac{1}{\sqrt{x_3 + 1}}$$

$$x_4 = \frac{1}{\sqrt{0.7576 + 1}}$$

$$x_4 = 0.7542$$

$$\epsilon_{\text{error}} = 0.01$$

$$\epsilon_{\text{error}} = \left| \text{Previous value} - \text{Current Value} \right|$$

$$\epsilon_{\text{error}} = |0.7419 - 0.7576|$$

$$\epsilon_{\text{error}} = |-0.01|$$

$$\epsilon_{\text{error}} = 0.01$$

lower upper triangle



Date _____

Solve by LU factorization.

5 Marks

10 min.

$$x_1 + 2x_2 + 5x_3 = -11$$

$$x_2 = 1$$

$$2x_1 + 9x_2 + 11x_3 = -20$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 1 \\ -20 \end{bmatrix}$$

Original Matrix :-

~~column row~~ row x column

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} & L_{21}U_{13} + U_{23} \\ L_{31}U_{11} & L_{31}U_{12} + L_{32}U_{22} & L_{31}U_{13} + L_{32}U_{23} + U_{33} \end{bmatrix}$$

$$U_{11} = 1, U_{12} = 2, U_{13} = 5$$

$$L_{21}U_{11} = 0 \quad L_{21}U_{12} + U_{22} = 1 \quad L_{21}U_{13} + U_{23} = 0$$

$$L_{21} \times 1 = 0 \quad 0 \times 2 + U_{22} = 1 \quad 0 \times 5 + U_{23} = 0$$

$$\boxed{L_{21} = 0}$$

$$\boxed{U_{22} = 1}$$

$$\boxed{U_{23} = 0}$$

$$L_{31}U_{11} = 2 \quad L_{31}U_{12} + L_{32}U_{22} = 9 \quad L_{31}U_{13} + L_{32}U_{23} + U_{33} = 11$$

$$L_{31} \times 1 = 2$$

$$2 \times 2 + L_{32} \times 1 = 9$$

$$2 \times 5 + 5 \times 0 + U_{33} = 11$$

$$\boxed{L_{31} = 2}$$

$$\boxed{L_{32} = 5}$$

$$\boxed{U_{33} = 1}$$

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$$\begin{bmatrix} 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} 0 \times 1 \\ 0 \times 1 \end{bmatrix}$$

Replaced in original matrix:-

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Apply First Step:-

$Ly = b \rightarrow$ Equation

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 1 \\ -20 \end{bmatrix}$$

$$y_1 = -11$$

$$y_2 = 1$$

$$2y_1 + 5y_2 + y_3 = -20$$

$$2(-11) + 5(1) + y_3 = -20$$

$$-22 + 5 + y_3 = -20$$

$$y_3 = -20 + 17$$

$$\boxed{y_3 = -3}$$

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Apply

Second step:-

$$Ux = y$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 1 \\ -3 \end{bmatrix}$$

$$x_1 + 2x_2 + 5x_3 = -11$$

$$\boxed{x_2 = 1}$$

$$\boxed{x_3 = -3}$$

$$x_1 + 2(1) + 5(-3) = -11$$

$$x_1 + 2 - 15 = -11$$

$$x_1 = -11 + 13$$

$$\boxed{x_1 = 2}$$

$$\text{Solution Set} = \{ 2, 1, -3 \}$$

Jacobian Matrix

Date _____

$$f_1 = x^2 + 2xy + y^2$$

$$f_2 = x^3 + 3y^2$$

$$\text{at } (x, y) = (2, 4)$$

$$J(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$$

$$f_1 = \sin^3 \theta + \cos \omega + \cos \phi$$

$$f_2 = \cos^3 \omega$$

$$f_3 = \sin^2 \phi \cos \omega$$

$$J(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial \theta} & \frac{\partial f_1}{\partial \omega} & \frac{\partial f_1}{\partial \phi} \\ \frac{\partial f_2}{\partial \theta} & \frac{\partial f_2}{\partial \omega} & \frac{\partial f_2}{\partial \phi} \\ \frac{\partial f_3}{\partial \theta} & \frac{\partial f_3}{\partial \omega} & \frac{\partial f_3}{\partial \phi} \end{bmatrix}$$

$$J(x, y) = \begin{bmatrix} 3\sin^2 \theta \cdot \cos \theta & -\sin \omega & -\sin \phi \\ 0 & -3\cos^2 \omega (\sin \omega) & 0 \\ 0 & \sin^2 \phi (-\sin \omega) & 2\sin \phi \cos \phi \cos \omega \end{bmatrix}$$

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$$J(x, y) = \begin{bmatrix} 2x+2y & 2x+2y \\ 3x^2 & 6y \end{bmatrix} \text{ at } (x, y) = (2, 4)$$

$$\Rightarrow J(2, 4) = \begin{bmatrix} 2(2)+2(4) & 2(2)+2(4) \\ 3(2)^2 & 6(4) \end{bmatrix}$$

$$\Rightarrow J(2, 4) = \begin{bmatrix} 4+8 & 4+8 \\ 12 & 24 \end{bmatrix}$$

$$\Rightarrow J(2, 4) = \begin{bmatrix} 12 & 12 \\ 12 & 24 \end{bmatrix}$$

Q: Find the Jacobian Matrix $f(x, y, z)$ of order 3×3 at the $(1, 3, 2)$ and $(3, 2, 1)$ for

$$f_1(x, y, z) = x^3 - y^2 + y - z^4 + z^2$$

$$f_2(x, y, z) = xy + yz + xz$$

$$f_3(x, y, z) = y/xz$$

$$\frac{\partial}{\partial y} \left(\frac{y}{xz} \right) = \frac{1}{xz}$$

$$\frac{\partial}{\partial z} \left(\frac{y}{xz} \right) \Rightarrow \frac{y}{x} \cdot \frac{1}{z} \Rightarrow \frac{y}{x} \cdot \frac{-1}{z^2} = -\frac{y}{x \cdot z^2}$$

$$J(x, y, z) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{bmatrix}$$

$$J(x, y, z) = \begin{bmatrix} 3x^2 & -2y+1 & -4z^3+2z \\ y+z & x+z & y+x \\ -\frac{y}{xz^2} & \frac{1}{xz} & -\frac{y}{xz^2} \end{bmatrix}$$

$$J(1, 3, 2) = \begin{bmatrix} 3(1)^2 & -2(3)+1 & -4(2)^3+2(2) \\ 3+2 & 1+2 & 3+1 \\ \frac{-3}{(1)^2(2)} & \frac{1}{(1)(2)} & \frac{-3}{(1)(2)^2} \end{bmatrix}$$

$$J(1, 3, 2) = \begin{bmatrix} 3 & -5 & -28 \\ 5 & 3 & 4 \\ -\frac{3}{2} & \frac{1}{2} & -\frac{3}{4} \end{bmatrix}$$

$$J(3, 2, 1) = \begin{bmatrix} 3(3)^2 & -2(2)+1 & -4(1)^3+2(1) \\ 2+1 & 3+1 & 2+3 \\ \frac{-2}{(3)^2(1)} & \frac{1}{(3)(1)} & \frac{-2}{(3)(1)^2} \end{bmatrix}$$

$$\Rightarrow J(3, 2, 1) = \begin{bmatrix} 27 & -3 & -2 \\ 3 & 4 & 5 \\ -\frac{2}{9} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

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CLO1 (Theory)

Date _____

~~Maximum~~

Differentiate between - Maximum and Minimum

- ~~First~~ If first derivative of function goes from positive to negative, then it is maximum.

~~If function's second deriv~~

- If first derivative of function goes from negative to positive, then it is minimum.

2nd derivative test :

if $y'' < 0 \Rightarrow$ maxima

if $y'' > 0 \Rightarrow$ minima

CLO-1 (Theory)

- Error
- Types of error
- Maximum / Minimum
- local Maximum / local Minimum

10
Approximate / 15 Number
Theory

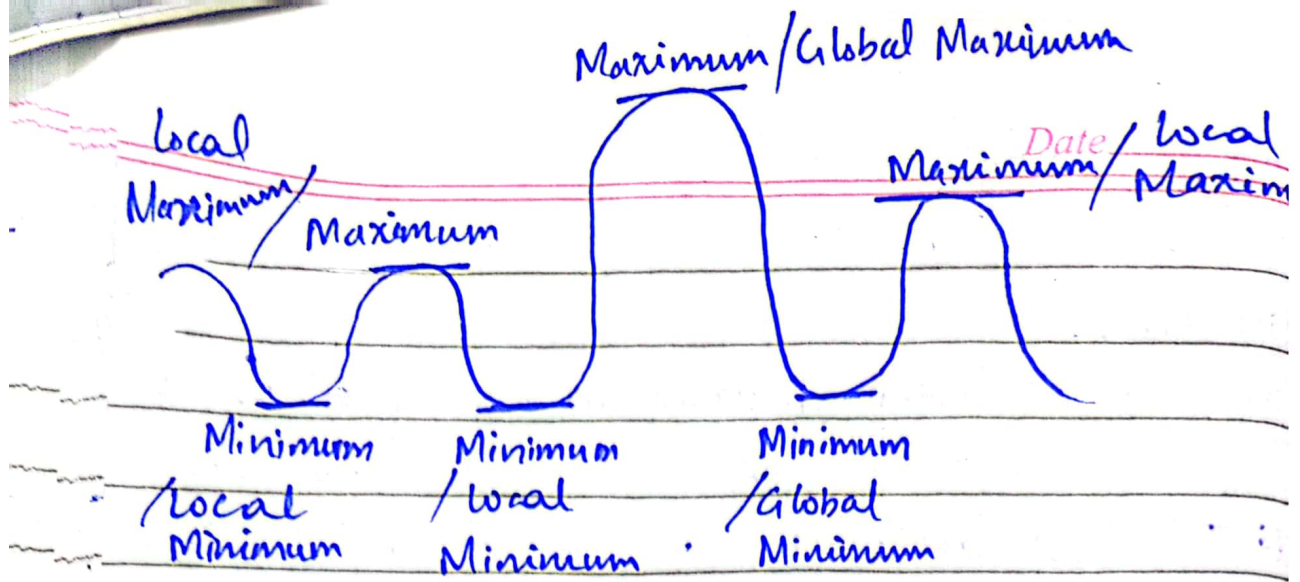
CLO-2

4 Question
Numerical

- 2 Non-Linear
Question (Any 2
out of 5 methods)

- 2 Linear
Questions (LU
factorization

and Jacobian
Matrix/Gauss
Elimination/Jordan
elimination



Local Minimum	Global Minimum	Local Maximum	Global Maximum
Smallest value within a neighborhood.	Largest Smallest Smallest value within over neighborhood the entire domain.	Largest value within a neighborhood.	Largest value over the entire domain.
Local region around a point.	Entire domain of the function.	Local region around a point.	Entire domain of the function.
Appears as a "valley" or dip.	Appears as the lowest point of the graph.	Appears as a "peak" or hill.	Appears as the highest point of the graph.
Diagram	Diagram	Diagram	Diagram

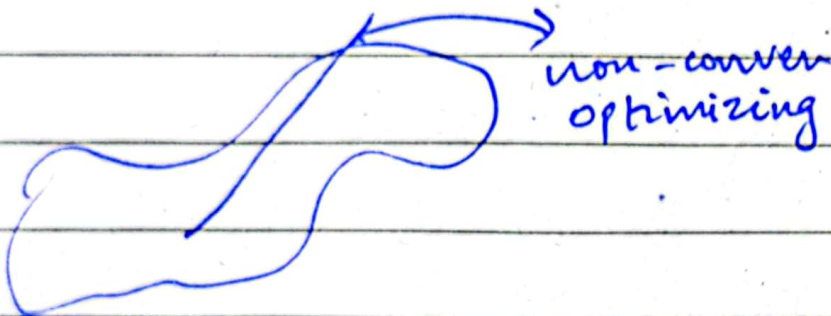
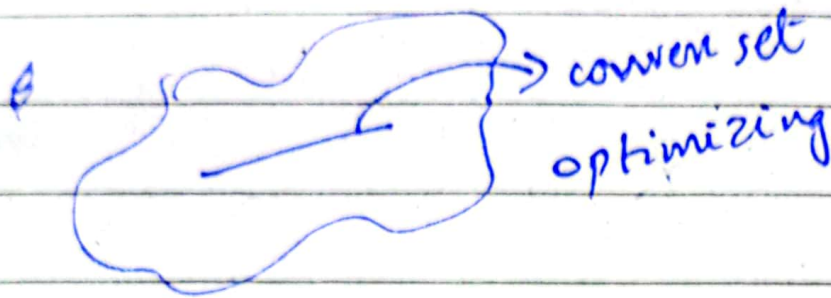
Support Vector

Machine: In binary classification, SVM aims to find the hyperplane that maximizes the margin between the two classes while minimizing classification errors.

Date: _____

Numerical optimizing is the process of finding the best solution to a problem within a given set of constraint.

* Difference between convex optimizing and non-convex optimizing



Convex Set Optimizing

- Portfolio Optimization
: linear relationship between the expected returns and the risk (covariance matrix).

Non-Convex Set Optimizing

- Neural Network Training:
Training deep neural networks involves optimizing a non-convex loss function with respect to the model parameters (weights and biases).

Least Square Regression:

To find the parameter
the sum of squared
difference between observed
value and predicted value.

Draw graph

Image Registration: In medical
imaging, image registration aims
to align different imaging modalities
or time series images.

Gradient Decent

Date _____

M	T	W	T	F	S	S
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First Order Optimization

Formula:-

$$X_{n+1} = X_n - (\text{learning rate}) \times \frac{dy}{dx}$$

Two value given:-

$$\alpha = 0.01$$

Initial Guess = 3
Value

01) $f(x) = x^2$

$$\Rightarrow \frac{dy}{dx} = 2x$$

$$x_1 = x_0 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_1 = 3 - (0.01)(2x)$$

$$x_1 = 3 - (0.01)(2(3))$$

$$x_1 = 3 - (0.01)(6)$$

$$x_1 = 2.94$$

$$x_2 = x_1 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_2 = 2.94 - (0.01)(2x)$$

$$x_2 = 2.94 - (0.01)(2(2.94))$$

$$x_2 = 2.94 - (0.01)(5.88)$$

$$x_2 = 2.88$$

$$x_2 = 2.88$$

$$x_2 = 2.88$$

$$x_3 = x_2 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_3 = 2.88 - (0.01)(2x)$$

$$x_3 = 2.88 - (0.01)(2(2.88))$$

$$\boxed{x_3 = 2.82}$$

$$x_4 = x_3 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_4 = 2.82 - (0.01)(2x)$$

$$x_4 = 2.82 - (0.01)(2(2.82))$$

$$x_4 = 2.82 - (0.01)(5.64)$$

$$\boxed{x_4 = 2.76}$$

$$\ln^5(\sin 2x)$$

$$5 \ln^4(\sin 2x)$$

$$Q2) f(x) = \sin^3(e^{x^2+2x})$$

$$Q3) f(x) = \ln^5(\sin 2x)$$

} Practice Question

$$Q4) f(x) = x^2 + 3x - 5$$

$$x_0 = 2$$

$$\alpha = 0.01$$

$$\Rightarrow \frac{dy}{dx} = 2x+3$$

$$\Rightarrow \text{Euler's } x_1 = x_0 - \alpha \left(\frac{dy}{dx} \right)$$

$$\Rightarrow x_1 = 2 - (0.01)(2x+3)$$

$$\Rightarrow x_1 = 2 - (0.01)(2(2)+3)$$

$$\Rightarrow x_1 = 2 - (0.01)(6+3)$$

$$\Rightarrow x_1 = 2 - 0.07$$

$$\Rightarrow \boxed{x_1 = 1.93}$$

$$x_2 = x_1 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_2 = 1.93 - (0.01)(2(1.93)+3)$$

$$x_2 = 1.93 - (0.01)(3.86+3)$$

$$x_2 = 1.93 - 0.0686$$

$$\boxed{x_2 = 1.86}$$

$$x_3 = x_2 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_3 = 1.86 - (0.01)(2(1.86)+3)$$

$$x_3 = 1.86 - (0.01)(3.72+3)$$

$$x_3 = 1.86 - 0.0672$$

$$\boxed{x_3 = 1.79}$$

$$x_4 = x_3 - \alpha \left(\frac{dy}{dx} \right)$$

$$x_4 = 1.79 - 0.01(2(1.79)+3)$$

$$x_4 = 1.79 - 0.01(3.58+3)$$

$$x_4 = 1.79 - 0.0658$$

$$\boxed{x_4 = 1.72}$$

5 6 Marks
Uses of Optimizing
Types of Optimizing
Uses of Types of Optimizing

Date

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Hessian Matrix $\rightarrow$ 2nd order optimization

Formula..

$$H \Rightarrow \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \text{ at } (1, 1)$$

$$f(x, y) = x^2y + y^2x$$

$$\frac{\partial f}{\partial x} = 2xy + y^2$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2x + 2y$$

$$f = x^2y + y^2x$$

$$\frac{\partial f}{\partial y} = x^2 + 2xy$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2x + 2y$$

Q: $f(x, y) = e^{y \ln x} \rightarrow$ Practice Question

$$f(x, y) = e^{y \ln x}$$

$$\frac{\partial f}{\partial x} = e^{y \ln x} (y \ln x)$$

$$\boxed{\frac{\partial f}{\partial x} = y \ln x e^{y \ln x}}$$

$$\frac{\partial^2 f}{\partial x \partial y}$$